

TIME RESPONSE ANALYSIS

The transfer function $G(s)$ of any system will be a ratio with numerator and denominator appearing as polynomials in s . Some examples are given in table 1.

Table 1. Examples of transfer function

$G(s) = \frac{2}{s+2}$	$G(s) = \frac{2s}{s+2}$	$G(s) = \frac{s+1}{s+5}$	$G(s) = \frac{s+2}{(s+5)(s+1)}$	$G(s) = \frac{2s^2+1}{4s^3+s^2+5s+2}$
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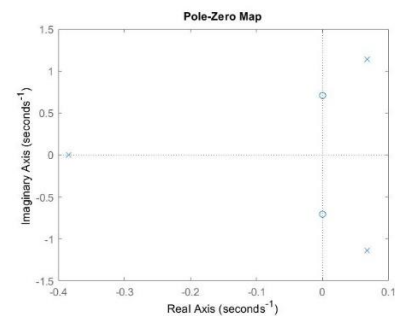
In general, the transfer function can be written as $G(s) = K \frac{p(s)}{q(s)}$ where K is the gain. The degree of $p(s)$ will be always less than the degree of $q(s)$. The equation $q(s) = 0$ is known as the **characteristic equation**, and the roots of the characteristic equation $q(s) = 0$ are known as the **poles**. The roots of the equation $p(s) = 0$ are known as **zeroes** of the system. The poles and zeroes of transfer functions given in Table 1 are shown below in Table 2.

Table 2. Poles and zeroes

Zero	No zero	$s = 0$	$s = -1$	$s = -2$	$s = 0 \pm j0.707$
Pole	$s = -2$	$s = -2$	$s = -5$	$s = -1, -5$	$s = -0.3842, 0.0671 \pm j1.1389$

The complex zeroes and poles appear as conjugate pairs.

The poles and zeroes can be represented in a complex plane (s is a complex variable $\sigma + j\omega$). The poles are represented by 'x' zeroes by 'o'. The pole-zero map of the last transfer function in Table 1 is shown in the figure.



#Exercise 1

Plot in Matlab/scilab, the Pole-Zero map of the transfer function

$$G(s) = \frac{2s^2+3s+5}{s(4s^3+s^2+5s+2)}$$

The poles and zeroes influence the transient and steady state responses of the system. We can always represent the transfer function as $G(s) = K \frac{\pi(s+z_i)}{\pi(s+p_i)}$ where K is the gain (constant), where K is the gain (constant), z_i represent the zeroes, p_i represents the poles, and π stands for the product. Hence the transfer function $G(s) = \frac{2s^2+1}{4s^3+s^2+5s+2}$ can be written as

$$G(s) = 0.5 \frac{(s+j0.707)(s-j0.707)}{(s+0.3842)(s-0.0671+j1.1389)(s-0.0671-j1.1389)}$$

The value of $K = 0.5$, zeroes $z_i = \pm j0.707$ and poles $p_i = -0.3842, 0.0671 \pm j1.1389$

Exercise 2

Using Matlab/scilab, plot in one figure the step response of the first order system by taking $a = 1, 3, \text{ and } 5$ and compare the time constants.

Exercise 3

Obtain the expression for the step response of $G(s) = \frac{b}{s+a}$. Compare the step responses of $G(s) = \frac{1}{s+2}$, $G(s) = \frac{2}{s+2}$, $G(s) = \frac{3}{s+2}$ by plotting them in the same figure using Matlab/Scilab

Ramp response

For a unit ramp, $x(t) = t$, $X(s) = \frac{1}{s^2}$

$$Y(s) = \frac{1}{s^2} \cdot \frac{a}{s+a} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+a} \quad \ll \gg \quad a = A s(s+a) + B(s+a) + C s^2$$

$$s = 0 \rightarrow B = 1 \quad \ll \gg \quad s = -a \rightarrow C = \frac{1}{a}$$

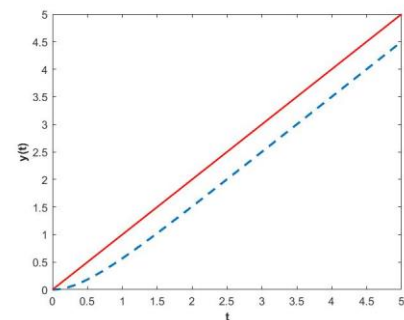
Equating coefficients of $s^2 \gg \gg 0 = A + C \gg \gg A = -C = -\frac{1}{a}$

$$Y(s) = \left(\frac{-1}{a}\right)\frac{1}{s} + \frac{1}{s^2} + \left(\frac{1}{a}\right)\frac{1}{s+a}$$

$$\therefore y(t) = \left(\frac{-1}{a}\right) + t + \left(\frac{1}{a}\right)e^{-at} = t - \frac{1}{a}(1 - e^{-at}) = t - T(1 - e^{-\frac{t}{T}})$$

The unit ramp response of a first order system $G(s) = \frac{2}{s+2}$ is shown in figure. Red line is the desired output(input) and blue line is the actual output.

As t takes a large value, the term $(1 - e^{-at}) \rightarrow 1$. Then $y(t) = t - \frac{1}{a}$. That is, at steady state, the actual output will be always less than the desired output by $(\frac{1}{a})$.



The error, $e(t) = x(t) - y(t) = t - \left[t - \frac{1}{a}(1 - e^{-at})\right] = \frac{1}{a}(1 - e^{-at})$

At steady state (as $t \rightarrow \infty$), the error $e(t) = \frac{1}{a}$

Impulse Response

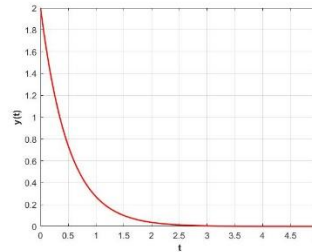
For a unit impulse, $x(t) = \delta(t)$, $X(s) = 1$

$$Y(s) = 1 \cdot \left(\frac{a}{s+a} \right)$$

$$y(t) = ae^{-at} = \frac{1}{T} e^{-\frac{t}{T}}$$

As $t \rightarrow \infty$, the response $y(t) = 0$

The unit impulse response of a first order system $G(s) = \frac{2}{s+2}$ is shown in figure. The input acts only for a short duration of time and as time progresses, the response exponentially decays to zero.



#Exercise 4

Using Matlab/Scilab, plot the impulse responses of the following first order systems in the same graph.

1. $G(s) = \frac{1}{s+1}$
2. $G(s) = \frac{5}{s+5}$

Now for a first order system with transfer function $G(s) = \frac{a}{s+a}$,

Ramp response $y(t) = t - \frac{1}{a}(1 - e^{-at})$

Step response $y(t) = 1 - e^{-at} = \frac{d}{dt}(\text{ramp response})$

Impulse response $y(t) = ae^{-at} = \frac{d}{dt}(\text{step response})$

#Exercise 5

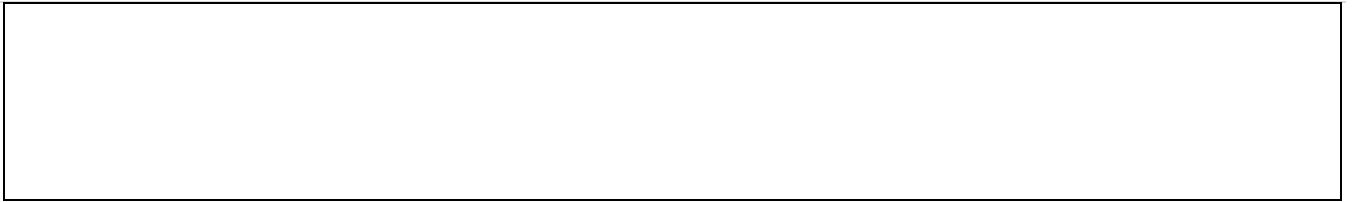
A first order system has a transfer function $G(s) = \frac{125}{100s+125}$. What is the time constant of the system? Determine the output of the system after $t = 0.5$ s when it is subjected to a unit step, unit ramp, and unit impulse inputs. What is the steady state error for unit ramp input?

#Exercise 6

A first order system has a unit step response given by $y(t) = 1 - e^{-0.75t}$. What will be its responses to unit ramp and unit impulse inputs after 1 s. What is the transfer function of the system?

References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons



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